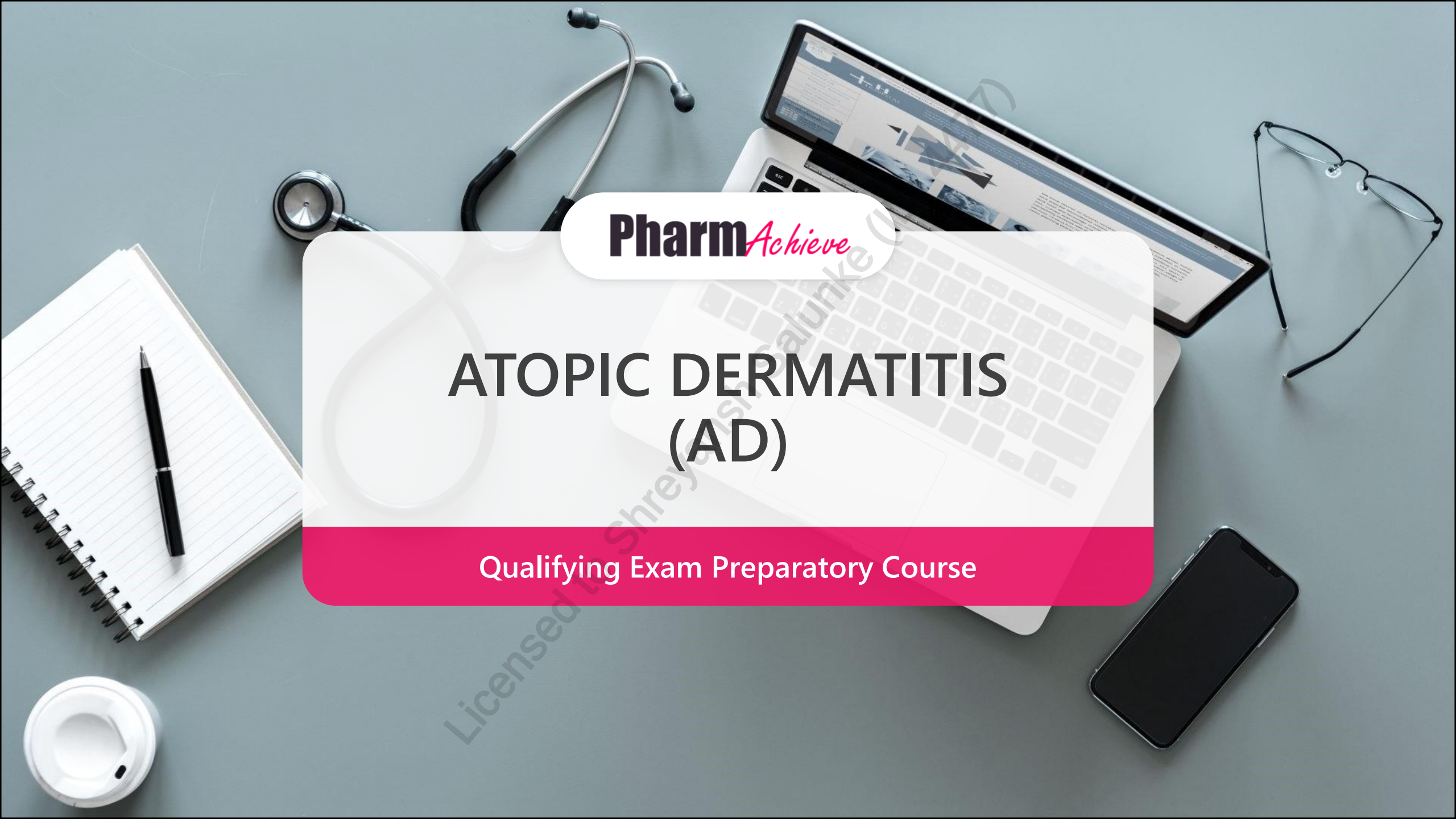


A top-down view of a grey desk with various items: a stethoscope, a laptop with a tablet on top, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

ATOPIC DERMATITIS (AD)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AD Atopic Dermatitis

BSA Body Surface Area

FTU Fingertip Unit

HPA Hypothalamic-Pituitary-Adrenal

JAK JAK Inhibitor

PDE Phosphodiesterase

TCI Topical Calcineurin Inhibitor

TCS Topical Corticosteroids

URTI Upper Respiratory Tract Infections

UV Ultraviolet

OBJECTIVES

- 1 Understand the pathophysiology of atopic dermatitis
- 2 Understand the clinical presentation and diagnosis of atopic dermatitis
- 3 Identify goals of therapy
- 4 Understand the management of atopic dermatitis, including non-pharmacological and pharmacological treatment options
- 5 Understand monitoring parameters

DEFINITIONS

Atopic dermatitis is an **inflammatory skin disease**

Eczema

General term used to define many forms of dermatitis

Atopic dermatitis

Atopic dermatitis is a **chronic, recurring** form of eczema

 the term “eczema” is frequently used for atopic dermatitis; however, there are **other skin conditions that are considered "eczemas" as well** (e.g. nummular eczema, seborrheic dermatitis and irritant contact dermatitis)

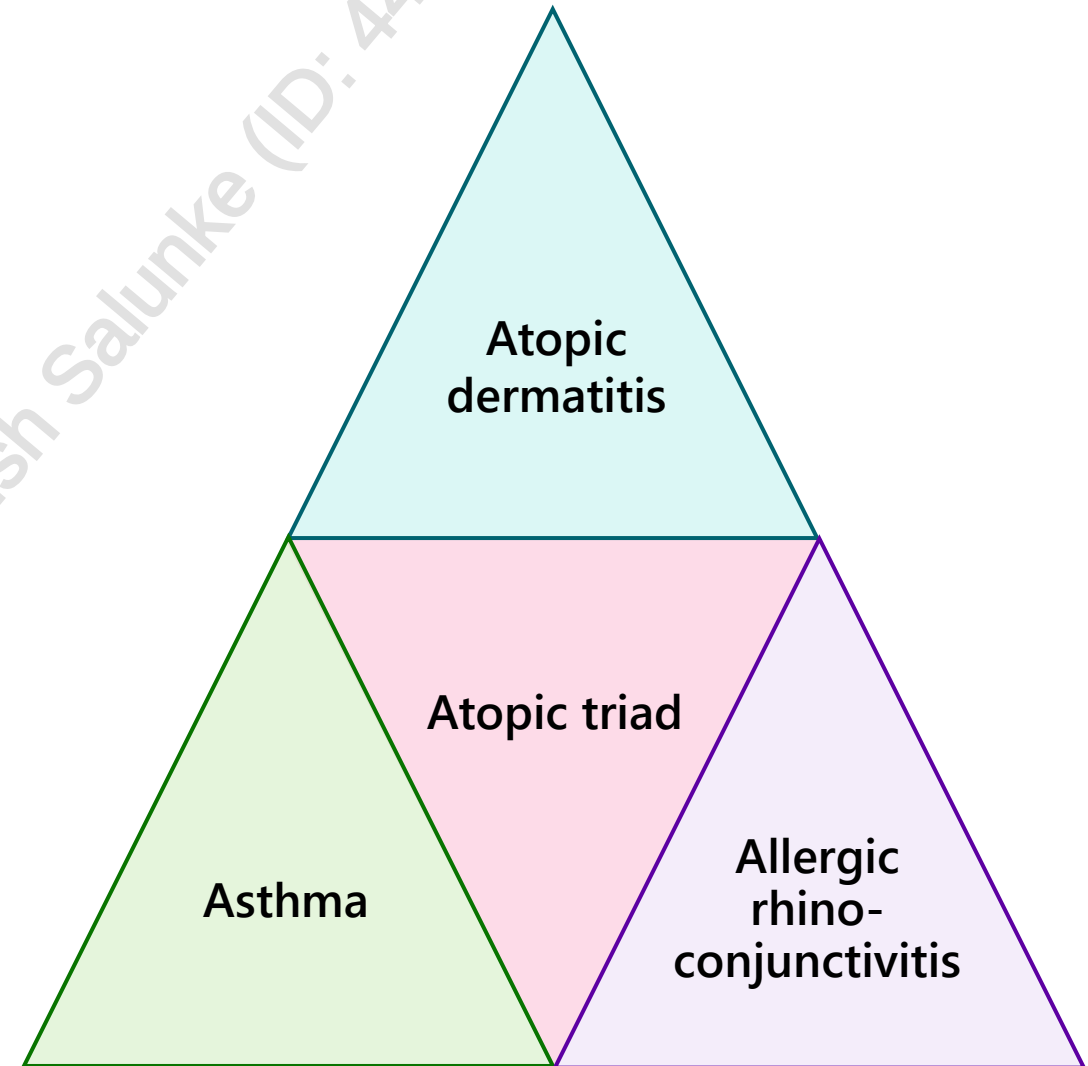
PATHOPHYSIOLOGY

Possible causes/risk factors

- Genetics
- Atopic triad
- Environmental exposures

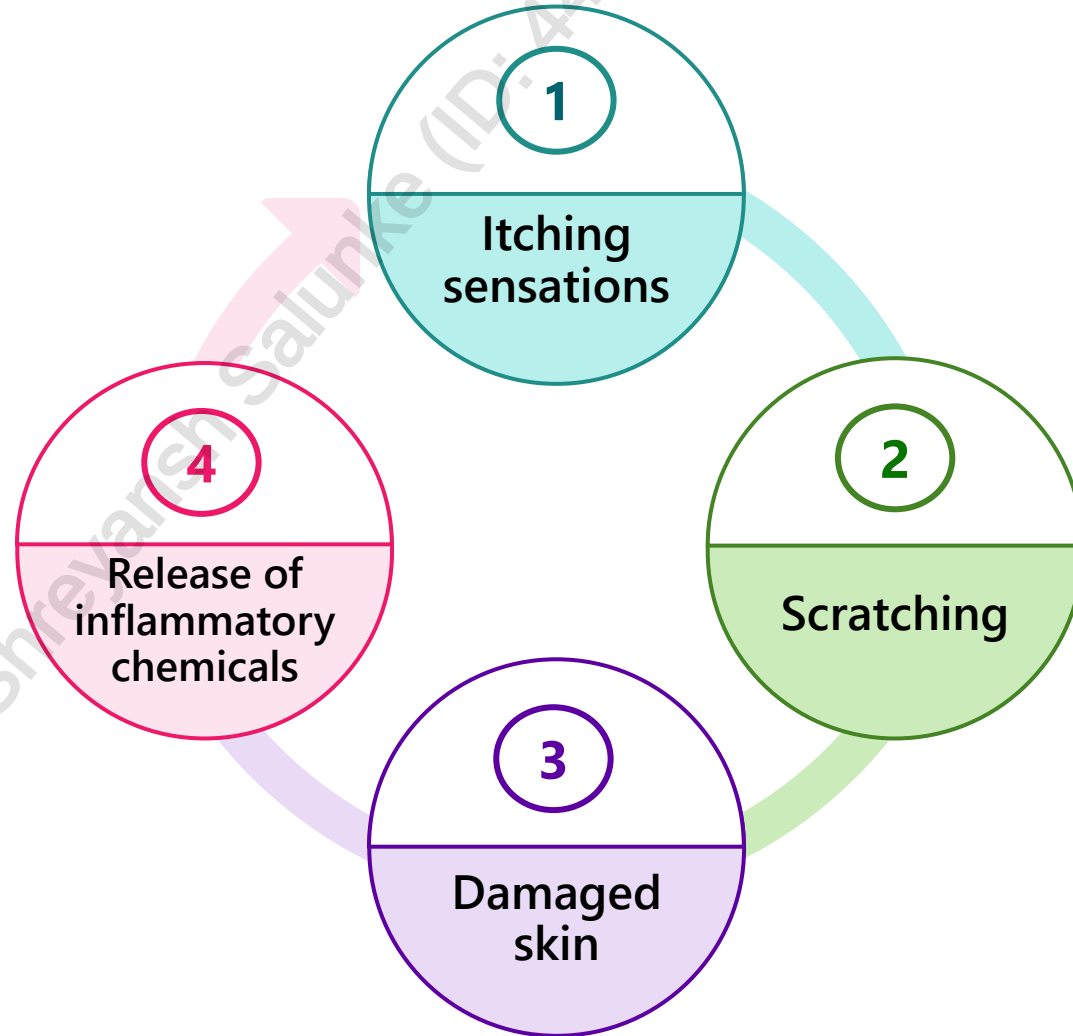
Possible triggers

- Allergens (e.g. mould, pollen, grass)
- Irritants (e.g. chemicals, fragrances, smoke)
- Fabrics (e.g. wool, synthetic fibres)
- Hot/cold temperatures
- Sweat/dry skin
- Skin infections
- Stress



CLINICAL PRESENTATION

- Dry/sensitive skin
- Burning skin sensations
- Skin inflammation
- Rash
- Scaly skin
- Redness
- Leathery skin patches
- Oozing
- Dark coloured skin areas



DIAGNOSIS

Physical assessments may indicate 1 of 3 typical morphologic patterns, but wide variations in distribution of rash

- Infants: facial and extensor dermatitis
- Older children: flexural and fold dermatitis
- Adults: facial and hand dermatitis

Bacterial swabs with moderate or heavy growth of organisms may suggest secondary bacterial infections in resistant patches

- Swabs containing minimal growth are difficult to interpret as many of the patients with atopic dermatitis are colonized with *Staphylococcus aureus*

GOALS OF THERAPY

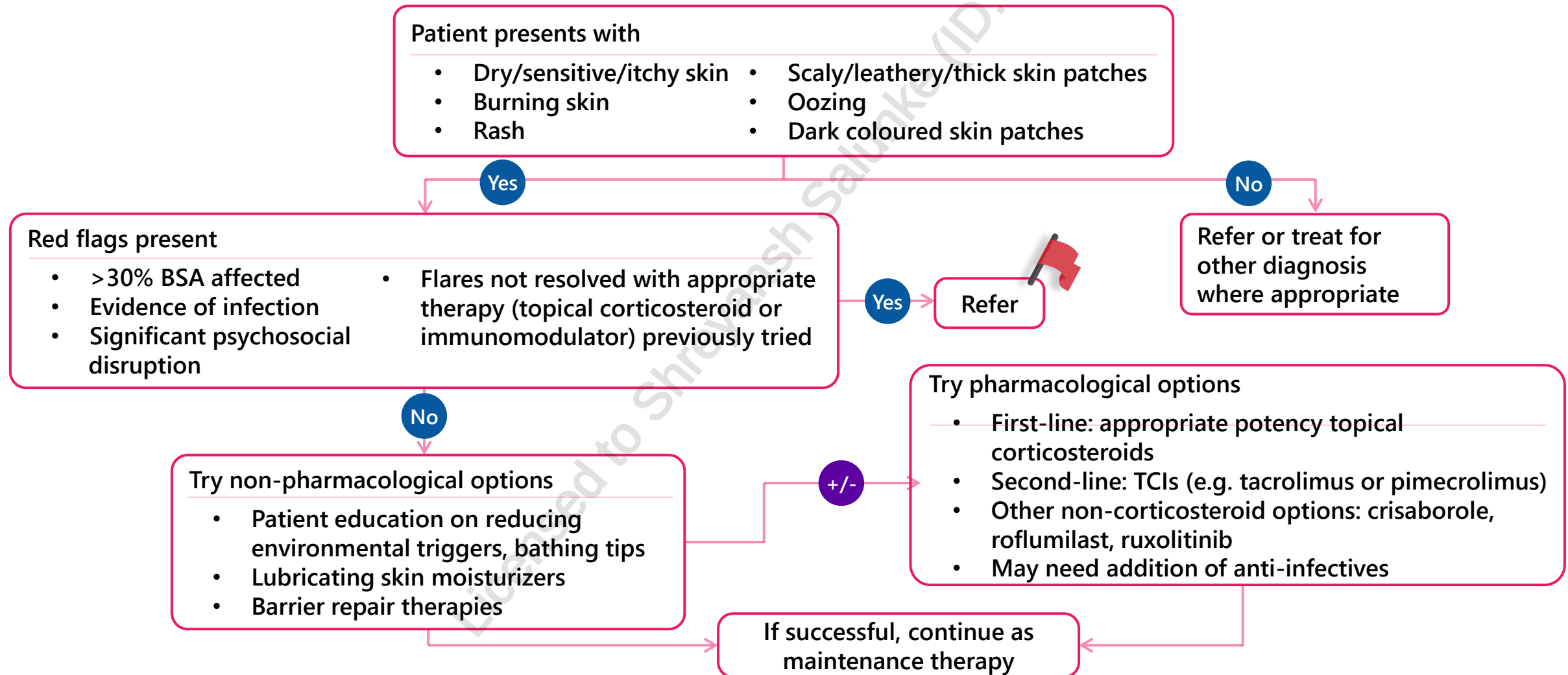
- ✓ Relieve symptoms, especially when they interfere with normal daily activities
- ✓ Treat patches of dermatitis to reduce inflammation and the risk of secondary infections
- ✓ Prevent flare-ups of symptoms resulting from environmental irritants
- ✓ Treat complications (e.g. secondary bacterial or viral infections)



Atopic dermatitis is a recurring chronic disorder that does not have a cure thus the major focus is to control symptoms

TREATMENT ALGORITHM

SUSPECTED ATOPIC DERMATITIS



NON-PHARMACOLOGICAL MEASURES

- Formal patient education programs
- Reducing environmental triggers: avoid perfumed products, dry grass and leaves, wool and synthetic fibres
- Keep comfortable temperature at home and work to minimize sweating
- Use **lubricating skin moisturizers** frequently and liberally
 - **First-line therapy** for preventing flare-ups and treating minimal irritation/itch
 - Should be used at least daily (even when drug therapy is required)
 - Should not contain irritants or fragrances
 - Plain petrolatum jelly is highly effective, non-irritating, and inexpensive
- **Advise patients:**
 - **Apply pharmacological treatments directly to the skin first → apply emollients on top**

NON-PHARMACOLOGICAL MEASURES

Wet wraps

- Wet bandages over moisturizers or medication
- May be helpful for extensive or recalcitrant AD
- Generally considered safe
- **Time-consuming and impractical**, proper education from an experienced physician is needed prior to use
- Risks of hypothermia, tissue maceration, infection, excessive absorption of medication

Bathing

- **No improvement of skin condition**
- Bathing should be brief (5–10 minutes)
- Use warm water
- Do not rub the skin aggressively when drying; pat the skin instead
- Apply moisturizers/emollients within 3 minutes of light drying

NON-PHARMACOLOGICAL MEASURES

BARRIER REPAIR THERAPIES

- **Barrier dysfunction:** disrupted ceramide content
 - Products **restore the correct balance of ceramides**
- Two studies revealed different results:
 - First study: Ceramide-dominant products are equivalent in efficacy to mid-potent corticosteroids after 28 days
 - Second study: When comparing ceramide-dominant products to another barrier repair therapy or to petrolatum-based moisturizers, there was no difference in managing mild-to-moderate eczema

PHARMACOLOGICAL MEASURES

Topical corticosteroids

Calcineurin inhibitors

Anti-infectives

Other

PHARMACOLOGICAL MEASURES

TOPICAL CORTICOSTEROIDS

- **First-line therapy for AD**
- Quick and effective by affecting several skin inflammatory pathways
- Available in a variety of vehicles and strengths
- “Steroid-phobia” often leaves patients with undertreated AD due to fears of systemic steroid side effects
 - Untreated AD may lead to infections, sleep loss, and decreased quality of life
- Continue until resolution of rash and itch, may be days-several weeks depending on patient
- Patients with flare-ups that do not significantly improve within 2-4 weeks should be reassessed

Skin Area and Characteristics	Corticosteroid Potency
Face and intertriginous folds: thin skin and greater absorption	Low potency
Body and scalp: medium skin thickness	Medium potency
Palms and soles: thick skin	High potency

PHARMACOLOGICAL MEASURES

TOPICAL CORTICOSTEROIDS

How can you choose the most appropriate therapy for your patients?

Vehicle	Pros	Cons
Cream	• Cosmetically elegant	• Less absorption • Additives may irritate
Lotion	• Evaporates well • Good for large surface areas • Good for hairy areas	• Alcohol base will irritate/sting open areas of AD
Gel	• Good for hairy areas • Good for oily skin	• Alcohol base will irritate/sting open areas of AD
Ointment	• Excellent penetration • Emollient effect • Little or no irritation	• Less acceptable cosmetically • Thick • Greasy

PHARMACOLOGICAL MEASURES

TOPICAL CORTICOSTEROIDS

FTUs FOR ADULTS

Area of skin to be treated	Size compared to volar surface of one hand (including fingers)	FTUs per application
One hand (front and back, including fingers)	2 hands	1
Face and neck	5 hands	2.5
One arm and hand	8 hands	4
One leg and foot	16 hands	8
Trunk (front)	14 hands	7
Trunk (back, including buttocks)	14 hands	7



1 Fingertip Unit (FTU) is the amount of product squeezed from a tube that reaches from the finger tip to the first crease of the finger. It should be used for dosing

PHARMACOLOGICAL MEASURES

TOPICAL CORTICOSTEROIDS

FTUs FOR CHILDREN (EVEN IN CHILDREN, FTU ARE MEASURED USING AN ADULT FINGER)

Area of skin to be treated	3-6 months	1-2 years	3-5 years	6-10 years
Face and neck	1	1.5	1.5	2
One arm and hand	1	1.5	2	2.5
One leg and foot	1.5	2	3	4.5
Trunk (front)	1	2	3	3.5
Trunk (back including buttocks)	1.5	3	3.5	5

PHARMACOLOGICAL MEASURES

TOPICAL CALCINEURIN INHIBITORS (TOPICAL IMMUNE MODULATORS)

- **Tacrolimus and pimecrolimus** -Second-line agents for short and long-term intermittent treatment
 - **Blocks calcineurin** through a **targeted anti-inflammatory mechanism**; in contrast to corticosteroids which have wide-ranging effects
 - **Significantly reduces severity** of eczema compared to placebo
 - Takes **longer to work** compared to corticosteroids
 - Although there are concerns about rare risk of malignancy (skin cancer, lymphoma), **a causal relationship has not been established with TCIs (black box warning was removed in Canada)**
 - **Off-label first-line therapy** for patients with eczema of the face or folds if there are concerns about the amount or frequency of use of low-potency corticosteroids
 - Patients can continue using corticosteroids anywhere else on the body that is required while using calcineurin inhibitors on the face or folds

PHARMACOLOGICAL MEASURES

CALCINEURIN INHIBITORS (TOPICAL IMMUNOMODULATORS)

- Tacrolimus
 - Indicated for **moderate to severe eczema**
 - Tacrolimus 0.03% has been shown to have a greater efficacy than mild corticosteroids
 - Tacrolimus 0.1% does not show any difference in terms of efficacy in comparison to mid-potency corticosteroids
- Pimecrolimus
 - Pimecrolimus 1% is indicated for treating **mild to moderate eczema**
 - Pimecrolimus has not been compared to low-potency topical corticosteroids for mild eczema
 - Pimecrolimus 1% is not as effective as betamethasone valerate 0.1% in patients with moderate eczema
- **Topical corticosteroids with calcineurin inhibitors are not more efficacious than topical corticosteroid monotherapy**

PHARMACOLOGICAL MEASURES

ANTI-INFECTIVES

- Secondary infections are common due to damaged skin barrier
- Usually caused by *S. aureus* (76-100% of patients are colonized with *S. aureus*), but can also be viral (herpes simplex) known as eczema herpeticum or fungal (dermatophytes)
- A worsening condition or failure to respond to appropriate therapy can be a sign of infection
- Other symptoms include papules and honey crusting (bacterial); punched-out erosions, hemorrhagic crusts, and/or vesicles (herpes simplex); and symptoms consistent with dermatophyte infections (fungal)
- Topical antibacterials (for minor, localized cases): Fusidic acid, gentamicin
- Oral antibacterials (for extensive or refractory cases) or IV Cephalosporins (if severe)
- Antivirals: Oral acyclovir (IV if severe)
- Topical antifungals: clotrimazole, ketoconazole, miconazole, terbinafine, tolnaftate and undecylenic acid (avoid nystatin as it is less effective)

PHARMACOLOGICAL MEASURES

OTHER

- **Topical PDE-4 inhibitor crisaborole** is a new therapy that is steroid-free
 - It is approved for mild-moderate AD for adults and children ≥ 3 months
- **Alitretinoin** is used in adults with severe chronic hand eczema (refer to a dermatologist)
- **Oral corticosteroids** should not be used routinely in patients with AD
- **Roflumilast** 0.15% cream: PDE-4 inhibitor. Indicated for mild to moderate AD in those ≥ 6 years old
- **Ruxolitinib** 1.5% cream: Janus kinase (JAK) inhibitor. Indicated for mild to moderate AD in ≥ 12 years old when other topical treatments are ineffective or inappropriate
- **No evidence for the effectiveness of antihistamines** since the pruritus in AD is not histamine-mediated
 - First-generation agents (e.g. diphenhydramine, hydroxyzine) are helpful if itching interferes with sleep
- **Systemic therapies** (e.g. azathioprine, cyclosporine, methotrexate, dupilumab, Upadacitinib, tralokinumab, lebrikizumab) or **ultraviolet (UV) light** are used in moderate to severe dermatitis and in patients who show no improvement/cannot tolerate topical therapies

DRUG TABLE

Class	Drug	Dose	Side Effects	Comments
Corticosteroids, low-potency	<i>Desonide</i> 0.05%	BID–TID	• Striae, telangiectasia, atrophy, purpura • When used around the eye for longer periods of time, ocular side effects may rarely occur • Systemic effects include suppression of HPA axis although clinically relevant features are very rare	• Good for face, intertriginous areas • Safe and effective when used appropriately
	<i>Hydrocortisone</i> 1%, 2%, 2.5%			

DRUG TABLE

Class	Drug	Dose	Side Effects	Comments
Corticosteroids, medium-potency	<i>Beclomethasone dipropionate</i> 0.025%	Daily–BID	• Striae, telangiectasia, atrophy, purpura • When used around the eye for longer periods of time, ocular side effects may rarely occur • Systemic effects include suppression of HPA axis although clinically relevant	• Good for body areas • Safe and effective when used appropriately • Derma-Smoothe/FS® product contains peanut oil but not peanut protein
	<i>Betamethasone valerate</i> 0.05%, 0.1%			
	<i>Clobetasone butyrate</i> 0.05%	BID–TID		
	<i>Fluocinolone acetonide</i> 0.01%			
	<i>Hydrocortisone valerate</i> 0.2%	BID–TID		
	<i>Prednicarbate</i> 0.1%	BID		
	<i>Triamcinolone acetonide</i> 0.1%, 0.5%			

DRUG TABLE

Class	Drug	Dose	Side Effects	Comments
Corticosteroids high-potency	<i>Fluocinonide</i> 0.05%	BID–TID	• Striae, telangiectasia, atrophy, purpura • When used around the eye for longer periods of time, ocular side effects may rarely occur • Systemic effects include suppression of HPA axis although clinically relevant features are very rare	• Good for thick, lichenified plaques • Safe and effective when used appropriately
	<i>Mometasone furoate</i> 0.1%	Daily		
	<i>Amcinonide</i> 0.1%	BID–TID		
	<i>Betamethasone dipropionate</i> 0.025%, 0.05%	BID		
	<i>Desoximetasone</i> 0.05%, 0.25%			

DRUG TABLE

Class	Drug	Dose	Side Effects	Comments
Corticosteroids ultra-potent	<i>Betamethasone dipropionate glycol 0.05%</i>	BID	• Striae, telangiectasia, atrophy, purpura • When used around the eye for longer periods of time, ocular side effects may rarely occur • Systemic effects include suppression of HPA axis although clinically relevant features are very rare	• Good for palms and soles • Safe and effective when used appropriately
	<i>Clobetasol propionate 0.05%</i>			
	<i>Halobetasol propionate 0.05%</i>			

DRUG TABLE

Class	Drug	Dose	Side Effects	Comments
PDE-4 Inhibitor	<i>Crisaborole ointment</i> 2%	BID	Mild application-site reactions	Adults and children ≥ 3 months: approved for mild-moderate AD
Calcineurin Inhibitors	<i>Pimecrolimus cream</i> 1%	BID (Patients ≥ 3 months)	Transient burning sensations, skin tingling, pruritus at site of application	- For use as a second-line until skin clears - Not for use in patients who are immunocompromised - Apply a thin layer and avoid unnecessary UV exposure - Indicated for patients with moderate to severe atopic dermatitis
	<i>Tacrolimus ointment</i> 0.03%, 0.1%	Pediatric (> 2 years): 0.03% ointment BID Adult (≥ 16 years): 0.03% or 0.1% ointment BID		
PDE-4 Inhibitor	Roflumilast 0.15% cream	Once daily	Application site reactions, headache, nausea	Indicated for mild to moderate AD in those ≥ 6 years old
JAK inhibitor	Ruxolitinib 1.5% cream	BID	URTI, diarrhea, nasopharyngitis, local reactions	Mild-to-moderate AD in ≥ 12 years when condition is refractory to TCI and TCS

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Scaly/dry/erythematous skin	Significant improvement/resolution	Within a few days to 4 weeks (with TCS use)*
Pruritus		
Flare-ups	Minimal-none	Ongoing

* Non-TCS therapies may take up to 6 weeks for symptom resolution

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Adverse effects (e.g. local irritation, stinging/burning, atrophy, purpura)	Minimal-none	Duration of therapy

TAKEAWAY

- AD is a **recurring, chronic disorder** that can be managed but not cured
- Ointments and gels result in less irritating and more potent corticosteroid products than creams or lotions
- Sweating, stress, and overheating can increase the amount of itching
- **First-line non-pharmacological treatment:** lubricating skin moisturizers
- **First-line pharmacological treatment:** topical corticosteroids
- **Second-line treatment:** TCI (tacrolimus, pimecrolimus)
- **Alternative non-corticosteroid options:** crisaborole, roflumilast and ruxolitinib
- Pruritus in AD is not histamine-mediated, so antihistamine use is limited to treating allergic symptoms such as allergic conjunctivitis and the sedating effect of first-generation agents may help with sleep

CASE 1

MK is a 29-year-old female who presents to her family physician with dry, red skin on her arms and face. She is frustrated as she feels the need to constantly itch her skin to get any sort of relief. She feels that this interferes with her daily activities and is embarrassed by the constant itching. She comes to the pharmacy to ask your advice on how to manage these symptoms. Her past medical history is significant for asthma for which she takes Symbicort® (budesonide/formoterol) 200/6 mcg 1 inhalation twice daily and as needed. She has seasonal allergies and is drinks socially.

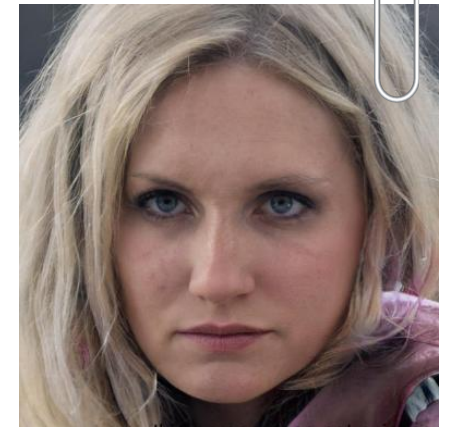
1. MK mentions that she uses oral and topical Benadryl® (diphenhydramine 25 mg) as needed during allergy season and is wondering if she even needs other therapy. Which of the following statements is NOT correct?
- a) Topical antihistamines have poor efficacy and may sensitize the skin
 - b) Topical corticosteroids are considered first line therapy for atopic dermatitis
 - c) Oral antihistamines are effective in treating atopic dermatitis
 - d) Oral diphenhydramine is helpful for sleep disturbances



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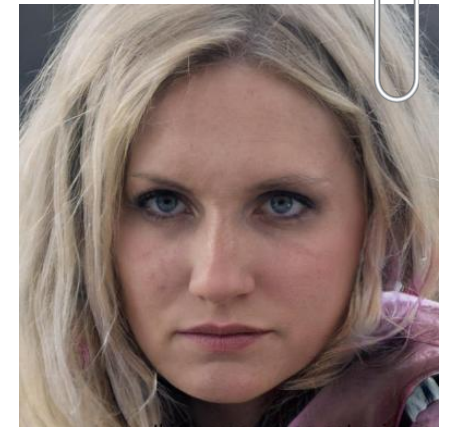
2. MK's physician decides to start her on a topical corticosteroid. Which of the following is true?
- a) She should not use a steroid as it can cause skin atrophy ("thinning" of the skin)
 - b) She should use a low potency steroid on her face and a medium potency steroid on her arms
 - c) She should use a low potency steroid on her arms and a medium potency steroid on her face
 - d) If her flare-ups do not improve within 1 week, she should be reassessed



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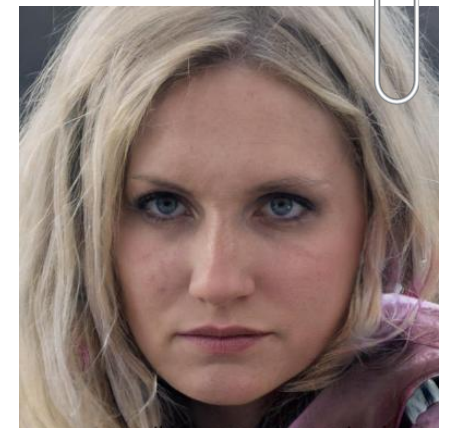
3. MK is fearful of using topical corticosteroids because she heard about steroid-induced skin atrophy and is looking for alternative options. Which of the following would be a suitable option?
- a) Apply a calcineurin inhibitor on her face once daily and a corticosteroid on her arms
 - b) Apply a calcineurin inhibitor on her face twice daily and a corticosteroid on her arms
 - c) Apply a calcineurin inhibitor on her face and her arms once daily
 - d) Switch to an oral corticosteroid



CASE 1

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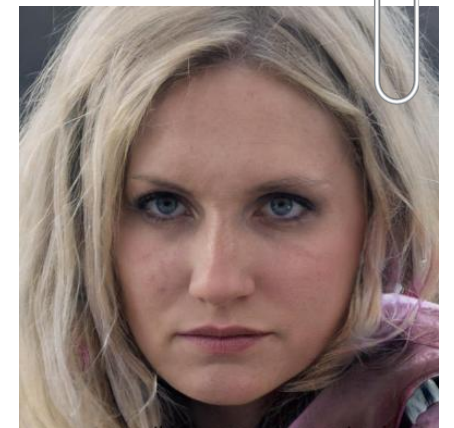
4. Which of the following is NOT a non-pharmacological measure to recommend to MK?
- a) Reduce exposure to triggers: environmental (e.g. grass, leaves), etc.
 - b) Use lubricating skin moisturizers when the patient feels very irritated
 - c) Bathing should be limited to 5-10 minutes
 - d) Emollients should be applied within 3 minutes after a bath or shower



CASE 1

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5. MK now presents with a prescription for hydrocortisone 3.2% ointment (mitte: 80 g), the pharmacy only has hydrocortisone 2% and 5% stock ointment available. How much of each ointment do you need to compound this prescription?
- a) 32 g of 2% and 48 g of 5%
 - b) 50 g of 2% and 30 g of 5%
 - c) 48 g of 2% and 32 g of 5%
 - d) 60 g of 2% and 20 g of 5%



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CHANGES LOG

March 2025

- Reformatted drug tables

August 2025

- Slide 3: Updated legend
- Slide 10: Edited pharmacological options in algorithm
- Slide 15: changed corticosteroid duration from 2 weeks to 2-4 weeks
- Slide 12: Added information to wet wraps
- Slide 19: Rephrased malignancy risk with TCI
- Slide 22: Added roflumilast and ruxolitinib according to the latest Canadian consensus published in April 2025
- Slide 22: Updated age for crisaborole use and edited information on antihistamine use
- Slide 27: Adjusted age for crisaborole use to ≥ 3 months and added roflumilast and ruxolitinib to drug table
- Slide 29: Updated Takeaway slide
- Slide 36: Reference list updated with new Canadian consensus guidelines

September 2025

- Slide 28: Elaborated on monitoring parameters to reflect new NAPRA competencies

October 2025

- Slide 16: Language change from eczema to AD
- Slide 22: Addition of tralokinumab, lebrikizumab for moderate to severe cases
- Slide 22: Addition of antihistamine examples (e.g. diphenhydramine, hydroxyzine)
- Slide 35: Updated references

February 2026

- Slide 28: Added note on timeline for non-TCS therapies